

Quick Recall: Key Definitions

Ordering (Ch. 2.1). A relation $\preceq$ on a set X is a (partial) *ordering* if it is reflexive, antisymmetric, and transitive. A (partial) ordering is a *linear ordering* if for every pair $x, y \in X$, either $x \preceq y$ or $y \preceq x$ holds. As a relation, $\preceq$ is a subset of $X \times X$.

Linear extension (Thm. 2.2.1). Let $(X, \preceq)$ be a finite partially ordered set. Then there exists a linear ordering $\leq$ on X such that $x \preceq y$ implies $x \leq y$. (Such $\leq$ is called a *linear extension* of $\preceq$.)

Complete bipartite graph (§4.7). For disjoint sets X and Y , $K_{X,Y}$ denotes the graph with vertex set $X \cup Y$

and edge set $\{\{x, y\} : x \in X, y \in Y\}$. Setting $a = |X|$, $b = |Y|$, this graph is isomorphic to $K_{a,b}$. It contains no triangle and has ab edges.

Turán's theorem (Thm. 4.7.1). For every n , $T(n) = \lfloor n^2/4 \rfloor$, where $T(n)$ is the maximum number of edges of a triangle-free graph on n vertices. Equality is attained uniquely (up to iso.) by $K_{\lfloor n/2 \rfloor, \lceil n/2 \rceil}$ (Thm. 4.7.2).

Edge-connectivity (§4.6). A graph G is *k-edge-connected* if we obtain a connected graph by deleting any $k - 1$ edges of G .

Practice: Textbook Theorem 4.7.3 — Degree-dominating bipartition for triangle-free graphs

Let $G = (V, E)$ be a triangle-free graph. Prove that there exists a partition of V into two subsets X and Y such that for every vertex $x \in V$,

$$\deg_G(x) \leq \deg_{K_{X,Y}}(x).$$

Remarks

One clever choice of X, Y does the whole job — no induction needed. Try:

- Pick $x_0 \in V$ of *maximum* degree in G . Let $Y = N_G(x_0)$ and $X = V \setminus Y$.
- For $x \in X$: what is $\deg_{K_{X,Y}}(x)$, and why is it $\geq \deg_G(x)$?
- For $y \in Y$: use that G is triangle-free to argue that Y is an *independent* set in G . What does this say about $N_G(y)$?

Corollary (Turán, Thm. 4.7.1). Summing the inequality gives $|E(G)| \leq |E(K_{X,Y})| \leq \lfloor n^2/4 \rfloor$.

Answer: Let $x_0 \in V$ be a vertex of maximum degree in G . Define

$$Y = N_G(x_0) = \{y \in V : \{x_0, y\} \in E\}, \quad X = V \setminus Y.$$

Note $x_0 \in X$ (since $x_0 \notin N_G(x_0)$). Also $|Y| = \deg_G(x_0)$.

Inequality for $x \in X$. Since $x \in X$, every neighbor of x in $K_{X,Y}$ lies in Y , so $\deg_{K_{X,Y}}(x) = |Y| = \deg_G(x_0)$. But x_0 was chosen of maximum degree, so

$$\deg_G(x) \leq \deg_G(x_0) = |Y| = \deg_{K_{X,Y}}(x). \checkmark$$

Inequality for $y \in Y$. We claim Y is an *independent set* in G : if $y_1, y_2 \in Y$ were adjacent, then $\{x_0, y_1, y_2\}$ would span a triangle in G (both y_i are neighbors of x_0 by definition of Y), contradicting the hypothesis.

So every neighbor of y in G lies outside Y , i.e. in X . Hence

$$\deg_G(y) = |N_G(y) \cap X| \leq |X| = \deg_{K_{X,Y}}(y). \checkmark$$

The partition (X, Y) satisfies the required inequality for every vertex. □

Corollary. Summing over all $v \in V$, $2|E(G)| = \sum_v \deg_G(v) \leq \sum_v \deg_{K_{X,Y}}(v) = 2|E(K_{X,Y})|$. Since $|E(K_{X,Y})| = |X| \cdot |Y| \leq \lfloor n^2/4 \rfloor$ (AM-GM / Ex. 4.7.1), we recover $|E(G)| \leq \lfloor n^2/4 \rfloor$.

Practice: Textbook Exercise 2.4.1(b)* — Partial orderings as intersections of linear orderings

Prove that every partial ordering $\preceq$ on a finite set X can be expressed as the intersection of linear orderings of X .

Remarks

Useful lemma (part (a) of Ex. 2.4.1). If $\leq_1, \dots, \leq_k$ are orderings on X , then $\bigcap_{i=1}^k \leq_i$ is again an ordering on X (check reflexivity, antisymmetry, transitivity from the definitions).

So the intersection of any family of linear extensions of $\preceq$ is automatically an ordering containing $\preceq$. The work is to choose *enough* linear extensions so that no extra pairs sneak into the intersection: for every incomparable pair $\{a, b\}$ in $\preceq$, build one linear extension with $a < b$ and another with $b < a$.

Answer: Write $\mathcal{I} = \{(a, b) : a, b \in X, a \neq b, a \not\preceq b, b \not\preceq a\}$ for the set of ordered pairs of *incomparable* elements. For each $(a, b) \in \mathcal{I}$, we build a linear extension $\leq_{(a,b)}$ of $\preceq$ with $a <_{(a,b)} b$ as follows.

Consider the relation $\preceq^{(a,b)}$ obtained from $\preceq$ by adding the pair (a, b) and taking the transitive closure:

$$x \preceq^{(a,b)} y \iff x \preceq y, \text{ or } x \preceq a \text{ and } b \preceq y.$$

This is still a partial ordering (reflexivity and transitivity are immediate; antisymmetry uses that $\preceq$ is an ordering and a, b are incomparable in $\preceq$ —no cycle is created). By Thm. 2.2.1, $\preceq^{(a,b)}$ has a linear extension $\leq_{(a,b)}$; note $a <_{(a,b)} b$.

Claim. $\preceq = \bigcap_{(a,b) \in \mathcal{I}} \leq_{(a,b)}$.

($\subseteq$) If $x \preceq y$, then $x \preceq^{(a,b)} y$ for every (a, b) , and hence $x \leq_{(a,b)} y$ for every (a, b) .

($\supseteq$) Suppose $x \neq y$ and $x \not\preceq y$. If $y \preceq x$, then $y \leq_{(a,b)} x$ for all (a, b) , so by antisymmetry $(x, y) \notin \leq_{(a,b)}$, and the intersection doesn't contain (x, y) . Otherwise x, y are incomparable in $\preceq$, and the linear extension $\leq_{(y,x)}$ puts $y <_{(y,x)} x$, so $(x, y) \notin \leq_{(y,x)}$, hence (x, y) is not in the intersection.

By part (a), this intersection is an ordering; the claim shows it equals $\preceq$. □

Remark. The infinite version is Szpilrajn's extension theorem (1930).

Practice: Textbook Exercise 4.6.6* — Toughness of d -regular, d -edge-connected graphs

Let $d \geq 3$ and let G be a d -regular (every vertex has degree d), d -edge-connected graph. Prove G is *tough*: for every $k \geq 1$ and every set $S \subseteq V(G)$ with $|S| = k$, the graph $G - S$ has at most k connected components.

Remarks

Count edges between S and the components of $G - S$. Each component contributes at least d edges to the cut (by d -edge-connectivity), but S only has dk endpoints available in total. Handle the small case carefully.

Answer: Let $S \subseteq V$ with $|S| = k$, and suppose $G - S$ has components $C_1, \dots, C_t$. We show $t \leq k$.

Case 1: $t = 1$. Then $t = 1 \leq k$. ✓

Case 2: $t \geq 2$. For each i , let e_i be the number of edges between C_i and S . Since $G - C_i$ is obtained from G by removing exactly the e_i edges of the cut $[C_i, V \setminus C_i]$, and $G - C_i$ is disconnected (the remaining components C_j for $j \neq i$ together with S form one side; wait—we need C_i separated from the rest). More carefully: the edges in the cut $[C_i, V \setminus C_i] = [C_i, S]$ (no edges go from C_i to C_j for $j \neq i$, since C_i, C_j are different components of $G - S$). So removing these e_i edges disconnects C_i from $S \cup \bigcup_{j \neq i} C_j$, which is nonempty (it contains $S \neq \emptyset$ since $k \geq 1$, and also C_j for $j \neq i$). Hence G minus these e_i edges is disconnected.

By d -edge-connectivity, $e_i \geq d$ for each i .

Now count edges between S and $V \setminus S$. Each vertex of S has degree d , so contributes at most d edges to this cut; summing,

$$\sum_{i=1}^t e_i \leq \sum_{v \in S} \deg_G(v) = dk.$$

Combined with $e_i \geq d$:

$$td \leq \sum_{i=1}^t e_i \leq dk, \quad \text{hence } t \leq k.$$

Remark. The hypothesis $d \geq 3$ is only needed to ensure G is nontrivial (for $d = 2$, cycles are 2-regular and 2-edge-connected but the bound $t \leq k$ still holds). The argument shows the stronger form: the number of components of $G - S$ is at most $|S|$ whenever the cut inequality $\sum e_i \leq d|S|$ combined with $e_i \geq d$ holds.